



A Level • Cambridge (CIE) • Chemistry

101 mins

54 questions

Multiple Choice Practice

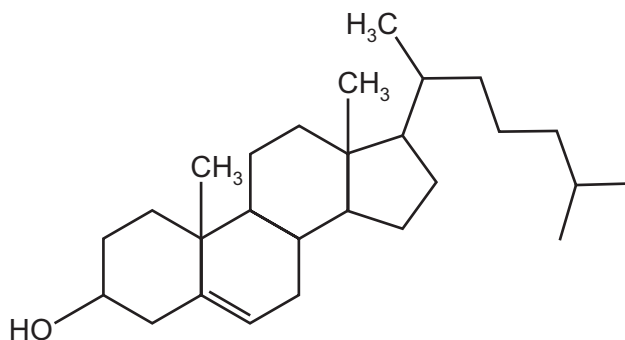
Topical Practice MCQ

13. An introduction to AS Level organic chemistry

Carefully selected past paper questions by topic.

25 The diagram shows the structure of the naturally occurring molecule cholesterol.

cholesterol



Student X stated that the 17 carbon atoms in the 4 rings all lie in the same plane.

Student Y stated that this molecule displays cis/trans isomerism at the C=C double bond.

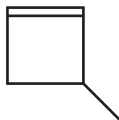
Which students are correct?

- A both student X and student Y
- B neither student X nor student Y
- C student X only
- D student Y only

- 25** Structural isomerism and stereoisomerism should be taken into account when answering this question.

The structure of 3-methylcyclobutene is shown.

3-methylcyclobutene



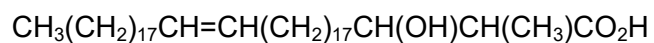
A mixture containing all stereoisomers of 3-methylcyclobutene is treated with HBr. This produces a mixture of isomeric bromomethylcyclobutanes.

How many stereoisomers does 3-methylcyclobutene have, and how many isomeric bromomethylcyclobutanes are present in the product mixture?

	3-methylcyclobutene	bromomethylcyclobutanes
A	2	4
B	2	6
C	4	4
D	4	8

- 26** Compound X is found in cell walls of some bacteria. Its structural formula is shown.

compound X



How many stereoisomers are there with this structural formula?

- A** 2 **B** 4 **C** 6 **D** 8

29 Cyclohexene, C_6H_{10} , is a hydrocarbon with a six-membered ring of carbon atoms.

It has several structural isomers that are straight-chain alkenes. The number of double bonds in each of these molecules is P.

What is the shape of the cyclohexene molecule and what is the value of P?

	shape	P
A	planar	1
B	planar	2
C	non-planar	1
D	non-planar	2

- 29 How many geometrical (cis/trans) isomers are there of hex-2,4-diene, $\text{CH}_3\text{CH}=\text{CHCH}=\text{CHCH}_3$?
- A none; hex-2,4-diene does **not** show geometric isomerism
 - B 2
 - C 3
 - D 4

26 In this question, alkenes and cyclic alkanes should be considered.

How many **structural** isomers of C_4H_8 are there?

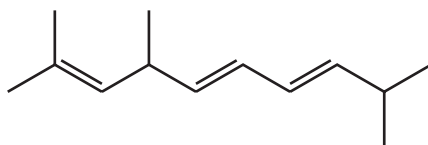
A 3

B 4

C 5

D 6

26 A skeletal formula is shown.



What is the total number of stereoisomers including the one shown?

A 4

B 6

C 8

D 16

28 Which pair of reagents react together in a redox reaction?

- A** $\text{CH}_3\text{CHCH}_2 + \text{Br}_2$
- B** $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + \text{concentrated H}_3\text{PO}_4$
- C** $\text{CH}_3\text{COCH}_3 + \text{HCN}$
- D** $\text{HCO}_2\text{C}_2\text{H}_5 + \text{dilute H}_2\text{SO}_4$

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- D $\text{HCO}_2\text{C}_2\text{H}_5 + \text{dilute H}_2\text{SO}_4$

35 Structural isomerism and stereoisomerism should be considered when answering this question.

How many isomeric compounds with molecular formula $C_5H_6O_4$ contain two $-CO_2H$ groups and one $C=C$ double bond?

A 5

B 6

C 7

D 8

22 Structural isomerism **and** stereoisomerism should be considered when answering this question.

A colourless liquid, $\text{C}_5\text{H}_{11}\text{Cl}$, exists as a mixture of two optical isomers.

When heated with sodium hydroxide in ethanol, a mixture of **only two** alkenes is formed.

What could the colourless liquid be?

- A** $(\text{CH}_3\text{CH}_2)_2\text{CHCl}$
- B** $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHClCH}_3$
- C** $\text{CH}_3\text{CH}_2\text{CCl}(\text{CH}_3)_2$
- D** $(\text{CH}_3)_2\text{CHCHClCH}_3$

22 Structural isomerism **and** stereoisomerism should be considered when answering this question.

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What could the colourless liquid be?

- A** $(CH_3CH_2)_2CHCl$
- B** $CH_3CH_2CH_2CHClCH_3$
- C** $CH_3CH_2CCl(CH_3)_2$
- D** $(CH_3)_2CHCHClCH_3$

25 Structural isomerism and stereoisomerism should be considered when answering this question.

How many non-cyclic isomers have the molecular formula C_5H_{10} ?

A 3

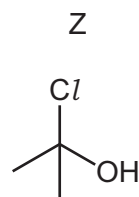
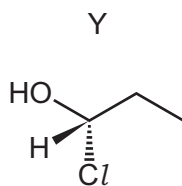
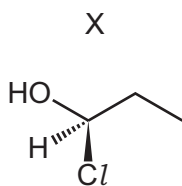
B 4

C 5

D 6

20 Structural and stereoisomerism should be considered when answering this question.

Compounds X, Y and Z are shown.



How many other isomers of C_3H_7ClO are there that are alcohols?

A 2

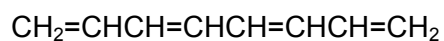
B 3

C 4

D 5

20 The unsaturated hydrocarbon octa-1,3,5,7-tetraene, C_8H_{10} , can display geometric isomerism.

octa-1,3,5,7-tetraene



How many isomers exist?

A 2

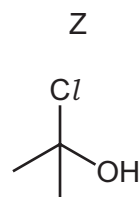
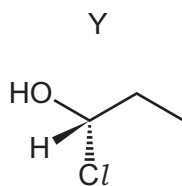
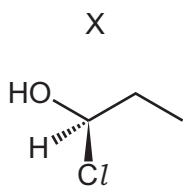
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C 4

D 8

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Compounds X, Y and Z are shown.



How many other isomers of $\text{C}_3\text{H}_7\text{ClO}$ are there that are alcohols?

A 2

B 3

C 4

D 5

20 How many structural isomers are there of trichloropropane, $C_3H_5Cl_3$?

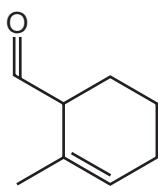
A 3

B 4

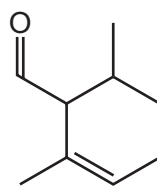
C 5

D 6

20 The diagrams show two different compounds.



1



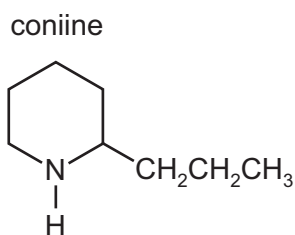
2

What is

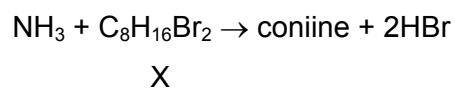
- the total number of structural isomers, including compound 2, that could be formed by adding a second methyl group to the ring of compound 1,
- the number of π electrons in each compound?

	number of isomers	number of π electrons
A	3	2
B	3	4
C	5	2
D	5	4

25 The structure of coniine is shown.



Coniine can be synthesised by reacting ammonia with a dibromo compound, X.



What is the name of compound X?

- A 1,1-dibromo-2-propylcyclopentane
- B 1,2-dibromo-2-propylcyclopentane
- C 1,4-dibromooctane
- D 1,5-dibromooctane

20 Structural isomerism and stereoisomerism should be considered when answering this question.

How many isomers with the formula C_5H_{10} have structures that involve π bonding?

A 3

B 4

C 5

D 6

20 Which types of stereoisomerism are shown by 2,4-dimethylhex-2-ene?

- A** both cis-trans isomerism and optical isomerism
- B** cis-trans isomerism only
- C** neither cis-trans isomerism nor optical isomerism
- D** optical isomerism only

24 Structural isomerism and stereoisomerism should be considered when answering this question.

A colourless liquid, $\text{C}_5\text{H}_{11}\text{Cl}$, exists as a mixture of two optical isomers.

When heated with sodium hydroxide in ethanol, a mixture of **only** two alkenes is formed.

What could the colourless liquid be?

- A $(\text{CH}_3\text{CH}_2)_2\text{CHCl}$
- B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHClCH}_3$
- C $(\text{CH}_3)_2\text{CHCHClCH}_3$
- D $\text{CH}_3\text{CH}_2\text{CCl}(\text{CH}_3)_2$

Question 23

9701_w18_qp_11 • Q21

- 21** X has the molecular formula $\text{C}_5\text{H}_{12}\text{O}$. X has a branched carbon skeleton and a secondary alcohol functional group.

How many **structural** isomers fit this description of X?

- A** 1 **B** 2 **C** 3 **D** 4

- 21 X has the molecular formula $C_5H_{12}O$. X has a branched carbon skeleton and a secondary alcohol functional group.

How many **structural** isomers fit this description of X?

- A 1 B 2 C 3 D 4

20 Structural isomerism and stereoisomerism should be considered when answering this question.

Each of the following carbonyl compounds is reacted with NaBH_4 . The product of each reaction is heated with Al_2O_3 at 600°C , generating one product or a mixture of isomers.

Which carbonyl compound will produce the most isomers?

- A** butanal
- B** butanone
- C** pentan-3-one
- D** propanone

21 Structural isomerism and stereoisomerism should be considered when answering this question.

If a molecule contains two non-identical chiral carbon atoms, four optical isomers exist.

How many isomers are there with

- molecular formula $C_7H_{14}O$ **and**
- a five-membered ring **and**
- a tertiary alcohol group?

A 4

B 5

C 9

D 13

30 How many esters with the molecular formula $\text{C}_5\text{H}_{10}\text{O}_2$ can be made by reacting a **primary** alcohol with a carboxylic acid?

A 4

B 5

C 6

D 8

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

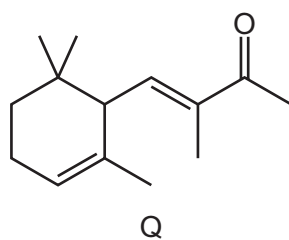
The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

Use of the Data Booklet may be appropriate for some questions.

20 The structural formula of compound Q is shown.



How many stereoisomers exist with this structural formula?

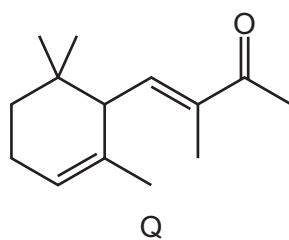
A 1

B 2

C 4

D 8

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How many stereoisomers exist with this structural formula?

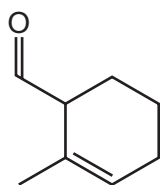
A 1

B 2

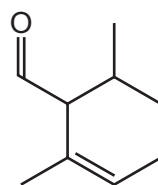
C 4

D 8

20 The diagrams show two different compounds.



1



2

What is

- the total number of structural isomers, including compound 2, that could be formed by adding a second methyl group to the ring of compound 1,
- the number of π electrons in each compound?

	number of isomers	number of π electrons
A	3	2
B	3	4
C	5	2
D	5	4

29 How many isomeric esters, including structural isomers and stereoisomers, can be made with the molecular formula $C_5H_{10}O_2$, if methanoic acid is one of the two reactants used?

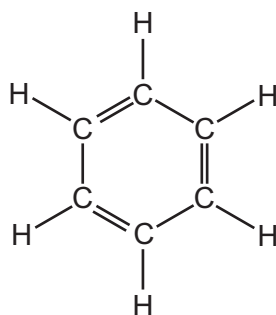
A 2

B 3

C 4

D 5

- 20 In 1865 Kekulé suggested a ring structure for benzene, C_6H_6 , in which a hydrogen atom is attached to each carbon atom.

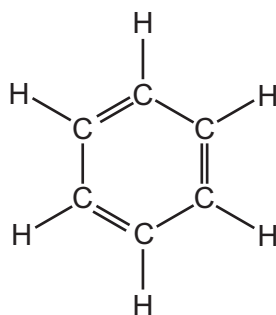


Kekulé structure

In this structure all of the bonds remain in the places shown. Assuming this is the structure of benzene, how many isomers of dichlorobenzene, $C_6H_4Cl_2$, would exist?

- A** 3 **B** 4 **C** 5 **D** 6

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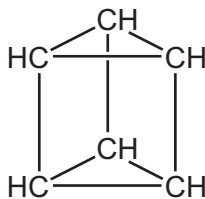


Kekulé structure

In this structure all of the bonds remain in the places shown. Assuming this is the structure of benzene, how many isomers of dichlorobenzene, $C_6H_4Cl_2$, would exist?

- A** 3 **B** 4 **C** 5 **D** 6

- 28 In 1869 Ladenburg suggested a structure for benzene, C_6H_6 , in which one hydrogen atom is attached to each carbon atom.



Ladenburg structure

A compound $C_6H_4Cl_2$ could be formed with **the same** carbon skeleton as the Ladenburg structure.

How many **structural** isomers would this compound have?

- A** 3 **B** 4 **C** 5 **D** 6

- 29 How many geometrical (cis-trans) isomers are there of hex-2,4-diene, $\text{CH}_3\text{CH}=\text{CHCH}=\text{CHCH}_3$?
- A none; hex-2,4-diene does not show geometric isomerism
 - B 2
 - C 3
 - D 4

20 In this question, structural isomerism **and** stereoisomerism should be considered.

How many isomeric aldehydes have the formula $\text{C}_5\text{H}_{10}\text{O}$?

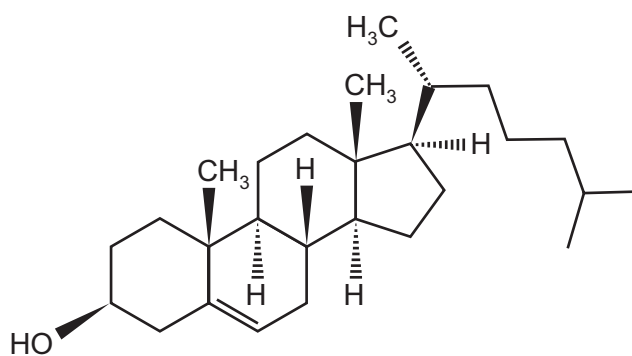
A 3

B 4

C 5

D 6

29 The diagram shows the structure of the naturally-occurring molecule cholesterol.



cholesterol

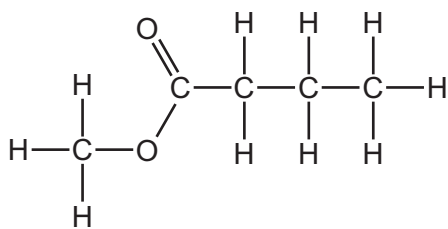
Student X stated that the seventeen carbon atoms in the four rings all lie in the same plane.

Student Y stated that this molecule displays *cis-trans* isomerism at the C=C double bond.

Which of the students are correct?

- A both X and Y
- B neither X nor Y
- C X only
- D Y only

- 30 Methyl butanoate, $C_5H_{10}O_2$, is an ester used in the food industry to give products the flavour of apples.



methyl butanoate

Including methyl butanoate, how many **structural** isomers are there of $C_5H_{10}O_2$ that are esters?

- A** 6 **B** 8 **C** 9 **D** 10

- 25 Pentane, C_5H_{12} , is reacted with chlorine in the presence of ultraviolet light. A compound R is found in the products. R has molecular formula $C_5H_{10}Cl_2$. Each molecule of R contains **one** chiral carbon atom.

Which two atoms of the pentane chain could be bonded to chlorine atoms in this isomer?

- A** 1 and 3 **B** 1 and 5 **C** 2 and 3 **D** 2 and 4

29 How many isomeric esters, including structural isomers and stereoisomers, can be made with the molecular formula $C_5H_{10}O_2$, if methanoic acid is one of the two reactants used?

A 2

B 3

C 4

D 5

23 How many isomeric esters have the molecular formula $\text{C}_4\text{H}_8\text{O}_2$?

A 2

B 3

C 4

D 5

21 An alkene has the formula $\text{CH}_3\text{CH}=\text{CRCH}_2\text{CH}_3$ and does **not** possess *cis-trans* isomers.

What is *R*?

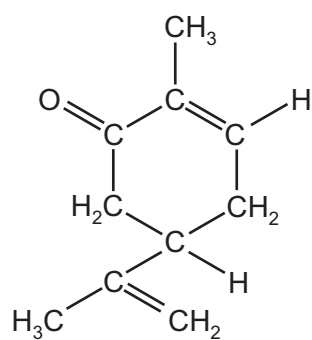
A H

B Cl

C CH_3

D C_2H_5

20 Carvone is found in spearmint.



carvone

How many σ and π bonds are present in this molecule?

	σ	π
A	13	3
B	22	3
C	22	6
D	25	3

21 An alkene has the formula $\text{CH}_3\text{CH}=\text{CRCH}_2\text{CH}_3$ and does **not** possess *cis-trans* isomers.

What is *R*?

A H

B Cl

C CH_3

D C_2H_5

23 Pentanol, $\text{C}_5\text{H}_{11}\text{OH}$, has four structural isomers that are primary alcohols.

How many of these primary alcohols contain a chiral carbon atom?

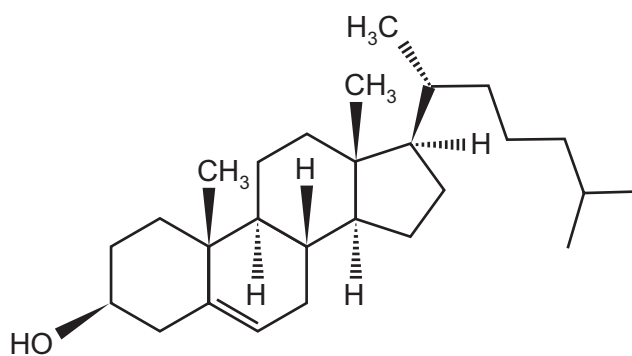
A 0

B 1

C 2

D 3

24 The diagram shows the structure of the naturally-occurring molecule cholesterol.



cholesterol

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Student Y claimed that this molecule displays *cis-trans* isomerism at the C=C double bond.

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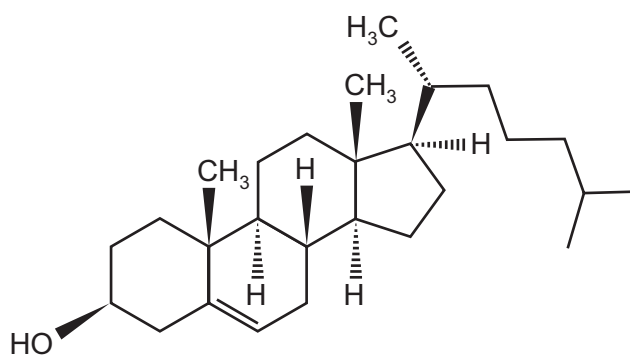
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cholesterol

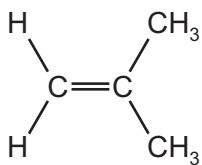
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- A both X and Y
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- C X only
- D Y only

- 29 The compound 2-methylpropene, C_4H_8 , is a monomer used in the production of synthetic rubber.



In addition to 2-methylpropene there are x other isomers of C_4H_8 , structural or otherwise, which contain a double bond.

What is the value of x ?

- A** 1 **B** 2 **C** 3 **D** 4

20 How many structural isomers are there of trichloropropane, $C_3H_5Cl_3$?

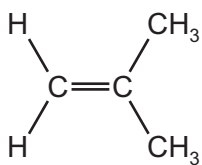
A 3

B 4

C 5

D 6

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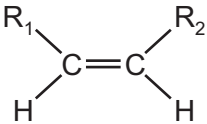
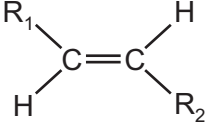
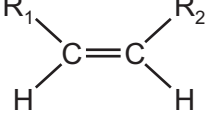
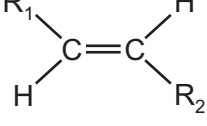
What is the value of x ?

- A** 1 **B** 2 **C** 3 **D** 4

- 21 The presence of 11-*cis* retinal, $C_{20}H_{28}O$, in cells in the eye is important for vision.

The structure of retinal includes an aldehyde group, a cyclohexene ring and a long aliphatic side chain, in which a carbon-carbon double bond exists between carbons numbered 11 and 12.

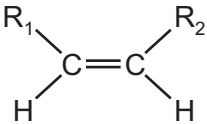
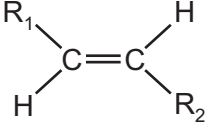
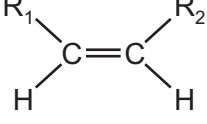
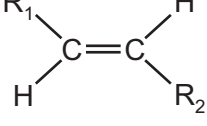
Which pair of statements about 11-*cis* retinal could be correct?

	total number of >C=C< double bonds	arrangement around the adjacent carbons 11 and 12
A	5	
B	5	
C	6	
D	6	

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The structure of retinal includes an aldehyde group, a cyclohexene ring and a long aliphatic side chain, in which a carbon-carbon double bond exists between carbons numbered 11 and 12.

Which pair of statements about 11-*cis* retinal could be correct?

	total number of >C=C< double bonds	arrangement around the adjacent carbons 11 and 12
A	5	
B	5	
C	6	
D	6	

Answer Key

No.	Paper	Topic	Answer
1	9701_m24_qp_12	13. An introduction to AS Level org...	B
2	9701_s24_qp_11	13. An introduction to AS Level org...	B
3	9701_w24_qp_12	13. An introduction to AS Level org...	D
4	9701_m23_qp_12	13. An introduction to AS Level org...	D
5	9701_w23_qp_12	13. An introduction to AS Level org...	C
6	9701_m22_qp_12	13. An introduction to AS Level org...	C
7	9701_s22_qp_12	13. An introduction to AS Level org...	C
8	9701_w22_qp_11	13. An introduction to AS Level org...	A
9	9701_w22_qp_13	13. An introduction to AS Level org...	A
10	9701_w22_qp_13	13. An introduction to AS Level org...	C
11	9701_w21_qp_11	13. An introduction to AS Level org...	D
12	9701_w21_qp_13	13. An introduction to AS Level org...	D
13	9701_m20_qp_12	13. An introduction to AS Level org...	D
14	9701_w20_qp_11	13. An introduction to AS Level org...	D
15	9701_w20_qp_12	13. An introduction to AS Level org...	B
16	9701_w20_qp_13	13. An introduction to AS Level org...	D
17	9701_m19_qp_12	13. An introduction to AS Level org...	C
18	9701_s19_qp_11	13. An introduction to AS Level org...	D
19	9701_s19_qp_11	13. An introduction to AS Level org...	D
20	9701_w19_qp_12	13. An introduction to AS Level org...	D
21	9701_m18_qp_12	13. An introduction to AS Level org...	D
22	9701_s18_qp_12	13. An introduction to AS Level org...	C
23	9701_w18_qp_11	13. An introduction to AS Level org...	A
24	9701_w18_qp_13	13. An introduction to AS Level org...	A
25	9701_s17_qp_11	13. An introduction to AS Level org...	B
26	9701_s17_qp_13	13. An introduction to AS Level org...	C
27	9701_s17_qp_13	13. An introduction to AS Level org...	C
28	9701_w17_qp_11	13. An introduction to AS Level org...	C

Answer Key

No.	Paper	Topic	Answer
29	9701_w17_qp_13	13. An introduction to AS Level org...	C
30	9701_s16_qp_11	13. An introduction to AS Level org...	D
31	9701_s16_qp_12	13. An introduction to AS Level org...	D
32	9701_w16_qp_11	13. An introduction to AS Level org...	B
33	9701_w16_qp_13	13. An introduction to AS Level org...	B
34	9701_s15_qp_11	13. An introduction to AS Level org...	A
35	9701_s14_qp_11	13. An introduction to AS Level org...	C
36	9701_w14_qp_13	13. An introduction to AS Level org...	C
37	9701_w14_qp_13	13. An introduction to AS Level org...	B
38	9701_w14_qp_13	13. An introduction to AS Level org...	C
39	9701_s13_qp_11	13. An introduction to AS Level org...	A
40	9701_s13_qp_13	13. An introduction to AS Level org...	D
41	9701_s12_qp_13	13. An introduction to AS Level org...	C
42	9701_w12_qp_11	13. An introduction to AS Level org...	D
43	9701_w12_qp_12	13. An introduction to AS Level org...	D
44	9701_w12_qp_12	13. An introduction to AS Level org...	D
45	9701_s11_qp_11	13. An introduction to AS Level org...	B
46	9701_s11_qp_11	13. An introduction to AS Level org...	B
47	9701_s11_qp_12	13. An introduction to AS Level org...	D
48	9701_s11_qp_13	13. An introduction to AS Level org...	B
49	9701_s11_qp_13	13. An introduction to AS Level org...	B
50	9701_w10_qp_11	13. An introduction to AS Level org...	C
51	9701_w10_qp_12	13. An introduction to AS Level org...	C
52	9701_w10_qp_13	13. An introduction to AS Level org...	C
53	9701_w09_qp_11	13. An introduction to AS Level org...	A
54	9701_w09_qp_12	13. An introduction to AS Level org...	A